

TFPT — Changelog

All changes and new results, by date

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This file is the canonical, dated record of every change to the TFPT theory development: new verification modules (vN), status changes, editorial passes and infrastructure. It is maintained *in the same change* as the work it records, and it is also imported at the end of the `introduction` document. Module numbers vN refer to `verification/vN_*.py`; claim IDs refer to rows of `verification/status_ledger.csv`.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289_marklocal_raw.py** (MARKLOCAL.RAW.01) decomposes the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — $\text{RP} + \text{gap} + 4 \text{ marks} \Rightarrow \text{curvature vanishes away from the marks}$; [E] L3 orbit-source (v195/v216), L4 Fourier-support on $4\mathbb{Z}$ (v264/v288), L5 full-operator $[\rho, \Lambda] = 0$ (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- L^2 QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk $\Rightarrow$ single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286_seam_equivalence_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ E_8 into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287_free_rp_bulk_to_holomorphic_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain $\text{RP} + \text{gap} + \text{chirality} + \text{Gaussian} + \text{invertible} \Rightarrow \mu=1 \Rightarrow \text{holomorphic} \Rightarrow (E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk $\Rightarrow$ single-sector boundary* (the AQFT form of $\det K=1$). 4/5 discharged.
- **v288_full_l2_subprincipal_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod{4}$, so $\Lambda = |n| + M_f$ commutes with $\rho = \text{diag}(i^n)$ on the full operator,

$\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$; generic curvature $\rightarrow 4.44$), lifting v201/v284 to L^2 . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284_route_i_rp_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP $\rightarrow$ DtN (v194), L2 marks $\rightarrow$ pillowcase (v195/v216/v214), L3 flat Steklov = $\sqrt{-\Delta_{\text{flat}}}$ (verified here), L4 covariance from DtN (v258), L5 μ_4 -invariance (v276/Lean/v280); [C] Troyanov ($\chi_{\text{orb}}=0 \Rightarrow$ unique flat $\tau=i$) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.
- **v285_route_ii_seam_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E] 3/4 discharged (no abelian sector $\Leftrightarrow \det K=1 \Leftrightarrow$ holomorphic $\Leftrightarrow E_8$; the tower $16 \rightarrow 4 \rightarrow 1$), and **its open lemma coincides with Route (i)'s** — both are “the raw seam is the $(E_8)_1$ -at- $\tau=i$ object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$. Build green, AUDIT OK.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280_pillowcase_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat $\tau=i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$ is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281_holomorphy_modular_data.py** (QGAMB.MODULAR.01): #anyons = $|\det \text{Gram}|$ ($E_8 \rightarrow 1$, $\text{SO}(16) \rightarrow 4$), the anyon-condensation tower $16 \rightarrow 4 \rightarrow 1$, Gauss–Milgram $\rightarrow c=8$; holomorphy ($\det K=1$) is the single Tier-B discriminator.
- **v282_e8_tau_i_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* — $\chi_{E_8}=j^{1/3}$ gives $\chi_{E_8}(i)=1728^{1/3}=12$, so $\tau=i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau=i$), **obligation count** $2 \rightarrow 1$.
- **v283_live_killtest_scorecard.py** (PRED.LIVE.01): the recent round’s computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay 1.55×10^{35} yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.03)

- **v279_qgeo_obligation_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat $\tau=i$ pillowcase metric* (Troyanov-unique, $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow$ mark-locality $\rightarrow E_8$. [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Troyanov + OS).
- **FORM.QGEO.03: the all-orders step is now machine-proved in Lean**. `SeamDeckClosure.lean` gains `diagonal_commutatesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\text{flat}})$

of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho=\text{diag}(i^n)$ — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- `v278_lszt_bridge_unitarity.py` (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity. [E] LSZ bridge: the EG T -products of S_{pert} (v269) are the OS Wick-rotated correlators (v240), so amputate+on-shell-residue gives $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; [C] the gap $\Delta=6\log(3/2)>0$ gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\text{Im } M = \sum \int d\Pi |M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector’s Stelle ghost (v269/rt_F) is non-unitary $\rightarrow$ QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- `v277_seam_calderon_e8_match.py` (QGAMB.TIERB.01) reduces Tier-B to ONE bit. [E] the full $(E_8)_1$ matching target is computed and matches: $c=8$, $\det \text{Cartan}(E_8)=1$, the holomorphic character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248=\dim E_8$ weight-1 currents (240 roots). [E] the discriminator is holomorphy — the $c=8$ counterexample $(D_8)=SO(16)_1$ has $\det \text{Cartan}=4$ (non-holomorphic), so the one condition is $|\det K|=1$. [C] the seam side matches via v276 (flat $\tau=i$) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- `v276_qgeo_flat_closes_commutator.py` (QGEO.SYM.03) collapses the bedrock to ONE geometric postulate. The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]=0$. [E] if $H = f(\Delta_{\text{flat}})$ and the order-4 deck $\rho (z \mapsto iz)$ is a flat- $\tau=i$ isometry, then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ *exactly*, to all orders (verified on the discretised square torus: $\rho^4=I$, $[\rho, \Delta]=0$, $[\rho, \sqrt{-\Delta}]=0$; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H . [C] $\tau=i$ is forced by order-4 ($j=1728$; hexagonal $j=0$ is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat $\tau=i$ pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt_F)

- `redteam/rt_F_qft4d.py` (REDTEAM.F.01) stress-tests the just-published round (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2 + M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the non-perturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor

over-determination is conditional on the Λ -branch readout (`LAMBDA.BRANCH.01`, [C]), the gravity route being unconditional (caveat added to `v274`). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, `v269–v275`)

- **Published to Zenodo.** Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton `v269`, the concrete one-loop quartic `v271` and gauge β -coefficients `v273`), the complete complex PMNS matrix and leptonic CP strength `v270`, the honest absolute ν -mass-scale typing `v272`, the over-determination of the single mass anchor `v274`, and the `QG.AMB.01` obligation roadmap `v275`. Suite 273 modules (highest ID `v275`; `v186/v226` skipped), Wolfram 271/271, `run_all.py` green, AUDIT OK, `npm` build clean. The Zenodo description (`zenodo_description.html`) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν -mass scale, EG gauge running, over-determination, QG roadmap, `v272–v275`)

- `v272_nu_mass_scale.py` (`FLAV.NUSCALE.01`) **types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the `v184` firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2 / M_R$ (one UV input, like v_{geo} , `v153`); [C] the observed $m_3 = 0.050$ eV with the TFPT PS→GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (`v249`) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] M_R is not a clean $\bar{M}_{\text{Pl}}$ power, so the absolute value stays open; [X] the NO floor $\Sigma m_\nu = 0.0586$ eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- `v273_eg_gauge_running.py` (`QFT4D.SPERT.03`) **derives the SM one-loop b_i via the EG gauge self-energy**. [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (`v271`), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same 41 as the carrier algebra, `v159`); [O] two-loop + all-order stay open. Exact b_i mirrored in Wolfram.
- `v274_scale_overdetermination.py` (`SCALE.OVERDET.01`) **shows the single mass anchor is over-determined**. [E]/[C] two independent routes to $\bar{M}_{\text{Pl}}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/4\pi^2) e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438 \times 10^{18}$ GeV, to 0.11%; the seam being a conformal $c=8$ CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens `v153/v78`.
- `v275_qgamb_roadmap.py` (`QGAMB.ROADMAP.01`) **gives the open gate QG.AMB.01 a module**. [E] Tier A gap decoupling ($\Delta_{\text{eff}} = 1.648 > 0$, `v36/v76`); [E]/[C] Tier B reduction to the holomorphic $(E_8)_1$ boundary net (`v77`); [C] the perturbative layer in hand (`v269/v271/v273`); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, `run_all` green, AUDIT OK, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, `v271`)

- `v271_eg_oneloop_quartic.py` (`QFT4D.SPERT.02`) **takes the `v269` S_{pert} skeleton from “exists” to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product T_2 ; the massless propagator has scaling degree $\text{sd}=2$ in $d=4$, the one-loop bubble has $\text{sd}=4=d$, so $\omega=\text{sd}-d=0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d, d)=1$, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly —

the *same* factor that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry $3 = s, t, u$ channels), EG initial condition $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$. [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Exact core mirrored in Wolfram (270/270). [E] build green, AUDIT OK.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **v270_pmns_jarlskog_assembly.py** (FLAV.PMNS.03) assembles the four TFPT channels ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, $\sin^2 \theta_{23} = \frac{1}{2}$, $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$, $\delta = 4\pi/3$) into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength. [E] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [C] the Jarlskog $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ ($J_{\text{max}} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (v88); [C] data: J_{max} vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026 , the negative CP-violating sign reproduced. [O] δ is the μ_6 /triality phase (assembled, not from M_ν/D_F) and the absolute ν -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269_spert_paqft_skeleton.py** (QFT4D.SPERT.01) builds the honest middle layer of the three-layer S-matrix (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers: S_{top} (2d DHR braiding $|M|=1$, statistics, v243), S_{pert} (4d Epstein–Glaser/BV S-matrix of the spectral action), S_{phys} (LSZ on the OS Wightman functions, v240). [E] the spectral-action a_4 interaction is power-counting renormalizable (all operators $\dim \leq 4$; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap $\Delta = 6 \log(3/2)$ gives an IR-safe adiabatic limit ($S_{\text{pert}} \rightarrow S_{\text{phys}}$ via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268_theta13_carrier_trace.py** (FLAV.TH13.01) closes the θ_{13} exponent origin. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); now the exponent is exact: [E] $\gamma = 5/6 = \text{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). [E] own channel: the $\mu\tau$ -breaking $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) “ θ_{13} from M_ν ” route flagged by the new script atlas. [O] the CP magnitude and the absolute ν -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make_script_atlas.py** generates **verification/script_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (**status_ledger.csv** for claim $\rightarrow$ status $\rightarrow$ dependencies $\rightarrow$ supersedes, **script_registry.csv/script_clusters.csv** for the theme grouping, **docs_map.csv** for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a

dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. [E] build green, AUDIT OK.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- `v267_qgeo_rigidity_minimal_axiom.py` (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks” ($\omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a ; cross-ratio 2 $\Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on $y^2=x^3-x$; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]=0$ and $\omega \circ \rho = \omega$ all follow (v214/v201/v264/v198). [E] neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal $j=0$ ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. [C] second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with $j=1728$ (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$) — the *reduction* is closed (axiom $\Rightarrow$ everything), the axiom is not. [E] build green, AUDIT OK.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- `v266_ps_threshold_proton.py` (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the v265 fork. The explicit two-step Pati–Salam thresholds give $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ($M_{GUT} \sim 2.4 \times 10^{15}$) gives $\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K and is *killed* even at the high hadronic band; [C] the E_8 -allowed $(15, 1, 1)$ [the single 45] raises $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$ yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ($\sim 1.4 \times 10^{35}$ yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be* $+(15, 1, 1)$), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- `v265_qft4d_fork_freeze.py` (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the $\alpha_1=\alpha_2$ scale α_3^{-1} off by ~ 5.6 , spread ~ 8.8 at 2×10^{16} GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the E_8 -allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2 \theta_W = 3/8$ (v248), the $O(1)$ $\kappa = M_{PS}/M_s$ band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake τ_p window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262_fqcd_mp_me.py (FR.MPME.02) implements the one contract-only F_{transfer} pipeline.** A real two-loop α_s solver runs $\alpha_s(M_Z)=0.1179$ with n_f thresholds (carrier $b_3=-7$, [E]) to $\alpha_s(2\text{ GeV})\sim 0.30$ and $\Lambda_{\overline{\text{MS}}}^{(3)}\sim 0.33\text{ GeV}$; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces $m_p/m_e\sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263_mnu_from_df.py (FLAV.PMNS.02) makes M_ν a seesaw operator of D_F .** [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$ -symmetric inputs) $\mu\tau$ -symmetric; [C] it reproduces $\theta_{23}=45^\circ$ and $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ (v9) under the seam misalignment; [O] $\theta_{13}=0$ in the leading texture, the CP *magnitude* (240°) is the μ_6 /triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **v264_raw_seam_marklocal.py (QGE0.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$ -invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGE0.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [E] AUDIT OK, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ closed modulo QGE0.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` “ambient closure” section now states that the cutoff of its S_{rel} is the seam KMS weight ($f_2/f_0=1$, v259) and D_{rel} the covariance induction (v258); the `tfpt_2` “(4) Constructive QFT closure” section adds the relative-object pointer; `tfpt_4_frontier`’s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (v255/v259); and `origin_theory`’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGE0.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary G_{net} entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGE0.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG

separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)

- **v258_dirac_covariance_induction.py** (PS.DIRAC.03) closes the “**posited Yukawa vs derived operator**” gap of v250/v252. The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, $\beta=1$) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$. [E] the inversion identity holds exactly; [E] C_F is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252 D_F (same mass spectrum); [E] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [C] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ of the boundary geometry and the v18 Yukawas are the readout of C_Σ .
- **v259_modular_cutoff_kappa.py** (PS.SPECACT.02) fixes the last open spectral-action input. The cutoff f is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes β , so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [E] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [E] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [E] hence $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$ loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa \approx 1.3$ pins $c_{PS}/c_{\text{grav}} \approx 1.7$).
- **v260_k3_kummer_unification.py** (ARCH.K3.01) unifies seam, carrier and lattice on one surface. The pillowcase is the elliptic-involution quotient $T_{\tau=i}^2/(z \mapsto -z)$ (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the E_8 Cartan is even/det 1/positive-definite; [E] $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1 , even, signature $(3, 19)$); [E] the seam is $T^2/(z \mapsto -z)$ with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the $|A[2]|=2^4=16$ Kummer nodes = $\dim S^+$ (v197), $16=4 \times 4$; [E] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [C] so the v1 “glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, v232/v236) and globally (H^2 summand). A unification, not a closure.
- **v261_modular_spectral_closure.py** (QFT.MSC.01) is the assembly capstone. The whole QFT round (v238–v260) is certified as *one* relative object $\text{TFPT}_{\text{QFT}} = (A_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to *one* open premise. [E] the cross-consistency invariants agree: $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6 \log(3/2)$ (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)

- The canonical changelog now also drives a public /changelog website page. A new generator `verification/make_changelog_web.py` parses this file (dated \subsection*

entries, `itemize` items, and the inline markup `\textbf/\emph/\texttt/\vref`, the status markers `[E]/[C]/[O]/[X]`, inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. `[E]` build (`npm run build green`, `/changelog` prerendered static) and `bash build.sh audit` (AUDIT OK).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257_quasiorient_modpd.py (PS.NCG.ORIENT.02) corrects the slightly too-generous v256 typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit v252 triple: `[E]` strict orientability fails ($\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$); `[E]` *quasi-orientability* holds (the left/right A -irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$); `[E]` strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- L/R -neutrino case); `[C]` modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254_inner_fluctuations.py (PS.NCG.FLUCT.01) closes v250 #5 + the no-junk obligation.** The inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator are computed explicitly: `[E]` they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1, 2)_{1/2}$ — no junk, no triplet, no $(10, 1, 3)$ /126. `[E]` the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and `[E]` σ is an SM singlet $(1, 1, 0)$ with $B-L = 2$. `[C]` $\sigma = \text{scalaron}$.
- **v255_spectral_action_expansion.py (PS.SPECACT.01) closes the spectral-action expansion (v252 #11).** The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: `[E]` structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl² + R^2 +Higgs-quartic, $N = 96$); `[E]` $b/a^2 = 1/3$ (top-dominated); `[E]` $\sin^2 \theta_W = 3/8$; `[E]` the R^2 spin-0 mode = the Starobinsky scalaron; `[C]` Higgs quartic $\rightarrow \sim 125$ GeV with σ ; `[C]` $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$ explicit; `[O]` exact decimal needs the cutoff moments.
- **v256_orientability_poincare.py (PS.NCG.ORIENT.01) — honest status, no faked proof.** KO-dimension 6 forces the intersection form $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be *antisymmetric*: `[E]` skew form (verified), `[E]` rank 2, corank 1, null $\sim (1, 1, -1)$ on the SM K-theory $\mathbb{Z}^3$; `[E]` the orientability local condition $[\gamma_F, \pi(a)] = 0$. `[C]` the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); `[O]` a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, v252–v253)

- **v252_full_finite_triple.py** (PS.DIRAC.02) advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma) — H_F$ the particle $\oplus$ antiparticle doubling of three $SO(10)$ 16-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ($J^2=+1$, $JD=DJ$), grading ($\gamma^2=1$, $[\gamma, a]=0$, $J\gamma=-\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}}=m_D^2/M_R$. [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.
- **v253_kappa_scalaron_ps.py** (PS.KAPPA.01) discharges residual 6(b) of v249. κ in $M_{PS} = \kappa M_s$ ($M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors $\times$ the two E_8 -allowed contents) to a tight $O(1)$ interval (≈ 1.3). [C] heat-kernel origin: $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [X] the 126-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ σ — the CCvS Pati–Salam mechanism, v251)

- **v251_first_order_sigma.py** (PS.DIRAC.01) advances the Dirac contract v250 from a contract to an exhibited mechanism. On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked ($J^2=+1$, $\gamma^2=1$, $D\gamma=-\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R \nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ . Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C] $\sigma = \text{scalaron}$ (v249); [O] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** v247_e8_branching_no126.py (PS.E8BRANCH.01): the E_8 hull branches under $SO(10) \times SU(4)$ as $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4})$, so the $SO(10)$ content is exactly $\{1, 10, 16, 45\}$ and the 126 is *absent* — the renormalisable 126_H is algebraically forbidden and $10+16+45$ is E_8 -supplied (only one 45 $\Rightarrow$ proton-marginal). v248_ps_algebra.py (PS.ALGEBRA.01): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$, exact hypercharges $Y = T_{3R} + (B-L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2 \theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). v249_ps_unification.py (PS.RGTEST.01): the two-step $PS \rightarrow SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0\text{--}1.7$. v250_dirac_triple_contract.py (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ($H_F = 3 \times 16$, the seven NCG axioms, $\sigma = \text{scalaron}$) that would turn “carrier is gauged” from a fork into a theorem. The root-level qft.txt carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **v246_unification_data_crosscheck.py (QFT4D.RGTEST.01)** — the first hard data confrontation of the spectral-action prediction. TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the E_8 cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}–10^{17}$ GeV (residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at 1.7×10^{14} GeV — the three never meet. Typed **[X]/[O]**: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (**figures/qft_skeleton.pdf**, **figures/qft_unification.pdf**) added via **make_figures.py** (PDF + website PNG) and embedded in **tfpt_research_contracts.tex**; a root-level **qft.txt** summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ 16 = one anomaly-free SM generation, v245)

- **v245_spectral_matter_content.py (QFT4D.MATTER.01)** discharges the computable core of the 4d contract v244. The carrier half-spinor (the $SO(10)$ 16) is verified, in exact rational arithmetic, to be one Standard-Model generation: $\mathbf{16} = SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions + ν_R), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$), the spectral-action normalisation gives $\sin^2 \theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3} g_1$), and the Higgs is the $(1, 2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of **CONTRACT.QFT4D.01** is now **[E]**; only the seam $\rightarrow$ Dirac realisation and the run-down couplings stay **[O]**. Wired into **run_all.py**, the ledger, the registry, cited in **tfpt_research_contracts.tex**, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238).** **v239_kms_thermal_time.py (DYN.KMS.01)**: the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. **v240_gns_os_reconstruction.py (DYN.GNS.01)**: GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{OS} = -\log T = -L$ is positive with gap Δ (the v64 mass gap). **v241_dhr_sectors.py (DHR.SECTORS.01)**: particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \rightarrow 4 \rightarrow 1$ (v235/v237). **v242_spin_statistics_modular.py (DHR.MODULAR.01)**: spin-statistics, the modular (S, T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. **v243_haag_ruelle_braiding.py (DHR.SMATRIX.01)**: the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. **v244_spectral_action_contract.py (CONTRACT.QFT4D.01)**: the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open **[O]** contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into **run_all.py**, the ledger and registry, cited in **tfpt_research_contracts.tex**, mirrored to the website.

2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New v238_modular_lindblad_dynamics.py** (DYN.SEMIGROUP.01, cluster registry). A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel’s generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time). [E] keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap Δ (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real ≤ 0 spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments** (`experiments/`, all `data_limited`, not load-bearing). From the `problem_b` black-hole-cosmology survey: (i) `ccbh-dark-energy` — the de Sitter seam interior ($w_{\text{in}}=-1$) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\text{in}} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) `gravastar-compactness` — the Nariai $Q_{\text{geom}}=3/8$ equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C}=3/8$ (shared de Sitter endpoint $1/2$); since $1/3 < 3/8 < 4/9 < 1/2$ this is a light-trapping ECO, yielding an echo template (round-trip delay ~ 0.7 ms at $62 M_\odot$, amplitude $\leq (2/3)^6$) — a [C] structural echo, no $\mathcal{C} \leftrightarrow Q_{\text{geom}}$ map proven. (iii) `cosmic-handedness` — a parity watchdog (Shamir JADES $\sim 3.3\sigma$ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- η_B — **full BDP Boltzmann ODE solve confirms the route at the frozen scale** (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`; `tfpt_4_frontier`, [C]). Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\text{CP}} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10} , with *no* free M_R dial. This moves the leptogenesis route from “solve pending” to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); `tfpt_4_frontier` Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically** (`experiments/recovery-channel`; `tfpt_horizon_readouts`, [C]; companion to v221). The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected $\lambda=1$ (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (`internal_consistency`, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain** (`experiments/gw-ringdown-echo`; `tfpt_horizon_readouts` search-targets). The

Stage-1 machinery (Kerr-ringdown subtraction $\rightarrow$ matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.

- **Scorecard + website synced.** `experiments/evidence_scorecard.json` now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website `EXPERIMENTS_AUDIT` mirror and the η_B prediction entry are updated to match. These are standalone `experiments/` surfaces (not in the verification suite or ledger); the `tfpt_4_frontier/tfpt_horizon_readouts` and `papers.ts/predictions.ts` edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, `infrastructure`).** The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with $c = 8$ ” ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$ glue realised at Lie level (`v143`) and its odd sectors twisted/Ramond ($E_8 = 120_{\text{NS}} + 128_{\text{R}}$, `v148`), while the $q(A_3)$ normalisation collapses into the one declared anchor (`v152`). Two exact adversarial checks added ($120+128 = 248$; glue index $= |\mu_4| = N_{\text{fam}}+1 = 4$); status stays `[C]/[O]` (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/ (`scripts/export-shadow.sh`, `verification/make_script_index.py`, `infrastructure`).** `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet (`v231_cp_mu6_phases`, `tfpt_5_redteam`, `[E]/[C]`).** Sharper than `v220/v225` (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the $j=0$ Eisenstein CM unit): δ_{CKM} leading $= \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$. The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the `v88` misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays `[C]` (`CP.MU6.01`).
- **The seam as the E_8 Kleinian singularity (`v232_e8_kleinian_seam`, `origin_theory`, `[E]/[C]`).** The du Val side of the McKay correspondence (`v219`) gives the open seam-realisation premise `QGE0.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\text{car}} + N_{\text{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan ($\det 1$, even, (-2) -curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1=0$, $\pi_1=2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val

exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8 ; TOP0.E8.01, bedrock stays [O]).

- **CP is the universal family/triality phase, only sheet-split (v233_cp_triality_phase, tfpt_5_redteam, [E]/[C]).** One step past v231 in the $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$ factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight). Since $\rho^4 = \rho^1 \rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (v234_seam_holomorphy_selection, tfpt_research_contracts, [E]/[O]).** The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing E_8 : (i) holomorphy (μ -index 1; Target A), (ii) the seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; v232), (iii) exactly one 1-dim irrep (v219). All equal $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8 ; and holomorphic $c=8=g_{\text{car}}+N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8 . So P2, G_{net} , Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk (v160)} \Rightarrow \text{holomorphic boundary} \Rightarrow E_8$; the single residual analytic step is “free bulk $\Rightarrow$ holomorphic boundary” (GATE.HOLO.01).
- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ (v235_seam_chern_simons, tfpt_research_contracts, [E]/[O]).** A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K -matrix with $\#\text{anyons} = |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|=1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ($\det 16$, 16 anyons) $\rightarrow SO(16)_1 = D_8$ ($\det 4$) $\rightarrow (E_8)_1$ ($\det 1$, holomorphic) — all rank 8, $c=8=g_{\text{car}}+N_{\text{fam}}$ — so holomorphic = E_8 = the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order- $|\mu_4|$ Lagrangian glue ($\det 16 \rightarrow 1$, vs. a partial order-2 isotropic $\rightarrow 4=D_8$). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue ($\det=1$)” = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem (GATE.HOLO.02).
- **The (2,3,5) Brieskorn singularity is the one generator of the skeleton (v236_brieskorn_capstone, origin_theory, [E]).** Capstone of the icosahedral round: the singularity $x^2+y^3+z^5$, whose exponents are the atoms ($|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$, has Milnor number $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ (a *fifth* origin of the “8”), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8 , link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, v233) is h^{10} in $\langle h \rangle = \mathbb{Z}/30$, the μ_4 clock (v223) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (TOP0.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam is SRE (v237_seam_sre_closure, tfpt_research_contracts, [E]/[O]).** Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the K -matrix ground-state degeneracy on a genus- g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so “no topological degeneracy on any closed surface” $\Leftrightarrow \det K=1$; among rank-8 even positive-definite K ($c=8=g_{\text{car}}+N_{\text{fam}}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2, 3, 5\}$ (v219_icosahedral_mckay, origin_theory, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \rightarrow \hat{E}_6$, $2O \rightarrow \hat{E}_7$, $2I \rightarrow \hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$ — it contains the 2, 3, 5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$, sum $30=h(E_8)$, sum of squares $120=|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex) (v222_cm_norm_duality, origin_theory, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, $j=1728$) gives $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3+2i)=13=\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j=0$) gives $N(3+2\omega)=7=\text{scalaron}$. The $(3, 2)$ split generates $(5, 6, 7, 13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).
- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (v223_coxeter_totative_clock, origin_theory, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7 \rangle = \{1, 7, 13, 19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h)=\text{rank}$ and $h=2 \cdot 3 \cdot 5$ (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227_degree_exponent_channel_split, tfpt_5_redteam, [E]).** Exponents sum $120=|R^+(E_8)|$ (magnitude channel), degrees sum $128=\text{rank} \cdot \dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228_rr_index_gate, origin_theory, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D=4=|\mu_4|$ on $\mathbb{P}^1$ forces $h^0=5=g_{\text{car}}$, $\text{rank } H_1=3=N_{\text{fam}}$, $h^0+H_1=8=\text{rank } E_8$, $\Lambda^{\text{even}}(\mathbb{C}^5)=16=\dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGE0.RR.01).
- **The seam as a finite recoverability code (v221_seam_qecc, origin_theory, [E]/[C]).** Code dimension $\dim S^+=16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1, (2/3)^6, (1/3)^6\}$, recovery bound $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220_cp_hexagonal_modulus, tfpt_5_redteam, [E]/[C]).** The seam deck is the square modulus $j=1728$ ($\mathbb{Z}/4$); its partner is the hexagonal $j=0$ ($\mathbb{Z}/6$, $\arg \rho = \pi/3$). The frozen $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$ has leading term $\pi/3 = \arg \rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (CP.MOD.01).
- **The dual normal frame (d, n) with CP as orientation (v225_dual_normal_frame, tfpt_5_redteam, [E]/[C]).** $d=a^\top R^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)=-\frac{1}{2}$ Nariai; $n=(5, -9, 6)$ reads $\det$ on the first-generation column; $\det(\mathbf{1}, d, n)=21=3 \cdot 7$; $\text{Im } \det(\mathbf{1}, d, \rho n)=21 \sin(\pi/3)$ — CP as orientation (DUAL.FRAME.01).
- **F_{transfer} as one path on the Sheet Diamond (v224_diamond_ftransfer_path, tfpt_2_standard_model, [E]/[C]).** $M(s, t)=R+Q \text{diag}(s, t, t)$: K, C, F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1, 8, 10), (1, 8, 16)$; the Koide source $\rightarrow$ pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229_lepton_frobenius_algebra, tfpt_2_standard_model, [E]/[C]).** $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$, product $\frac{32}{9}$; $A=\mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius

algebra over the μ_6 resolvent (LEP.FROB.01).

- **The center budget (7, 11, 13) as three local norms (v230_center_budget_norms, tfpt_2_standard_model, [E]).** $C=R+Q \text{ diag}(1, 0, 0)$ row sums (7, 11, 13): $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (hex), $13=N_{\mathbb{Z}[i]}(3+2i)$ (square), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218_diamond_axis_geometry, tfpt_2_standard_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C+xU) = 14+6x$ (slope $6=|R^+(A_3)|$) and quadratic on the sheet axis $\det(C+yV) = 14+14y+4y^2$ (2nd difference $8=\text{rank } E_8$); the anchor block has $\det B(C+xU) = 2 \det(C+xU)$ and sheet curvature $6=|R^+(A_3)|$ — two curvatures $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$ lifts in steps $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$ then $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$ — the decuple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2 \text{Disc}(q)$ with squares $(1, 3, 4, 6)$ and kernels $(13, 48, 65, 105)$, F carrying $105 = 3 \cdot 5 \cdot 7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28=2 \dim G_2$, $52=\dim F_4$ stay audit-only, not spine, as in v94/v95); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed (v216_marks_gauss_bonnet, tfpt_research_contracts, [E]; ledger QGEO.MARKS.03).** The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2, 3, 6)$, $(2, 4, 4)$, $(3, 3, 3)$, $(2, 2, 2, 2)$, and three independent criteria — all cone orders = 2 (the $|\mathbb{Z}_2|$ sheet branch), $\text{rank } H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2, 2, 2, 2)$. The numerical sibling v217_marks_emergence_scan ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension → 248/248); v217 numerical (Python-only). Suite → 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise (v215_seam_deck_killtest, tfpt_research_contracts, [E]/[O]; ledger QGEO.KILL.01).** A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4 -deck-invariant). Levers: **K1** exactly four marks ($b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio 2 $\Rightarrow j = 1728$, $\text{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0 \mathbb{Z}/6$; generic

$\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f \mathbb{Z}_4$ -invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side **[E]** (regression guards), with the *decisive* kill deferred to `QGEO.REALIZE.01` **[O]** (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via `v214/v54/v56`). Suite $\rightarrow$ 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (`v214_seam_pillowcase`, `tfpt_research_contracts`, **[E]/[O]**; ledger `QGEO.PILLOW.01`). The two separately-tracked open QGEO residuals — `QGEO.ISO.01` (`v180`: the carrier clock is an order-4 *isometry* of the seam metric) and `QGEO.SUBPRIN.01` (`v201`: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2, 2, 2, 2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the `v201` “flat away from the marks”. **(2)** New link: cross-ratio(μ_4) = 2 (`v168`) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$; all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* `v201` and `v180`, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close `QGEO.SYM.01`: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ j -invariant/CM arithmetic); Klein- J values `mpmath-numerical`. Suite $\rightarrow$ 214 modules.

2026-06-15 (the F_{transfer} transfer functor contract `CONTRACT.F.01`)

- F_{transfer} formalised as a typed functor with four axioms (`v213_ftransfer_functor`, `tfpt_research_contracts`, **[C]**). The four frontier transfers are consolidated into ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)** μ_4 -deck equivariance (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53 = a^T(R+Q)$, $\text{Pl}(F) = (1, 22, 30)$, $\|\text{Pl}(K)\|_1 = 11$); **(3)** positivity/stochasticity ($\text{spec } T$ positive, $\kappa_f \in (0, 1)$); **(4)** external modules explicit ($b_3 = -7$ a carrier output). The four instances are F_{pole} (`v183`), $F_{\text{Boltzmann}}$ (`v212`), F_{relic} (`v211`), F_{QCD} (`v164`). `CONTRACT.F.01` is the third research contract alongside (U_{wall}) and (G_{metric}), the structural complement of the `v187` typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger `CONTRACT.F.01`.

2026-06-15 (frontier: two sharper **[C]** scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch** $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (`v211_axion_spine_angle`, `tfpt_4_frontier`, **[C]**). A more *robust* misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (`v185`, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108^\circ$ exactly (no fit), the finite- T solver reaches Ω_{DM} at $\approx 106^\circ$, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1 - \varphi_\Sigma)$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); **[X]** $\Omega_a h^2$ outside $\sim [0.08, 0.16]$ demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decouple branch** (`v212_leptogenesis_decouple`, `tfpt_4_frontier`, **[C]**). Both Boltzmann inputs share the decouple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx$

8.65×10^9 GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than v184's $M_R \varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10} . The coupling mechanism is posited not derived, so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, [O]). A new module machine-proves the COHO node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4 - 1)$ ($k=1,2,3$) have pullback character $\rho^* \omega_k = i^k \omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since $i^4=1$), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3)$ = the A_3 exponents = $\text{Spec}(Q_+)$, rank 3 = N_{fam} . The MODULE *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^* \omega_1 = \omega_3$, $\sigma^* \omega_2 = \omega_2$, $\sigma^* \omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in AuditContract.lean. The geometric COHO/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (MobiusUniformisation.lean, ledger FORM.QGEO.02, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the clock $\rho : z \mapsto iz$ has $\rho^4 = \text{id}$ and $\rho^2 \neq \text{id}$ (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma \rho \sigma = \rho^{-1}$ (so $\langle \rho, \sigma \rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes 1, -1, swaps $i, -i$); and the multiplier classification “ $z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in AuditContract.lean. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation QGEO.MARKS.01 stays [O], *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega \circ \rho = \omega$ implication is now Lean-formalised** (SeamDeckClosure.lean, ledger FORM.QGEO.01, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no sorry/admit, no domain axioms — #print axioms = propext, Classical.choice, Quot.sound only): (i) the 4th-root character orthogonality $\sum_{j < 4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$ for $\zeta^4=1$ (so a μ_4 -mark sum is supported only on modes $\equiv 0 \pmod{4}$); (ii) the clock generator $-i$ has order 4; (iii) markLocal_blockDiagonal: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed structure SeamDeckPremise (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as CalderonProjector encodes the Paper-1 analytic input; (v) SeamDeckPremise.clock_invariant: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level** (v210_mark_local_dtn, tfpt_research_contracts, [O]). A numerical sharpening of QGEO.SUBPRIN.01/v201: a real von-Mises curvature bump summed over the four μ_4 marks,

$f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \text{sgn } H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close QGEO.SYM.01; net existence and full-cone RP remain the [E] of v175). Ledger QGEO.MARKLOCAL.01. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (v204_muon_seam_g2, tfpt_4_frontier, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ ; $\sim 1.5\sigma$ vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.
- **An independent gravitational $3/4$ (v205_xi_threequarter, tfpt_4_frontier, [E]/[C]).** $\xi = c_3/\varphi_{\text{tree}} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.
- **Quantitative H_0 from the Λ branch (v206_h0_lambda_branch, tfpt_4_frontier, [C]).** $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1/3}} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5\text{--}67.1$ km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route (v207_asymptotic_safety, tfpt_4_frontier, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric} . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam (v208_bh_thermodynamics, tfpt_horizon_readouts, [E]/[C]).** Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point (v209_bh_regular_core, tfpt_horizon_readouts, [C]).** No singularity ($\varphi \rightarrow \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (tfpt_3, [O]).** Added a paragraph to the “Möbius origin of π ” section: $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\text{PSL}(2, \mathbb{C})$ rotation (v180) and the RP $\rightarrow$ OS causal cone is the same Lorentz structure — so “Möbius origin of π ” and “one Lorentzian cone” are two faces of one boundary $\text{PSL}(2, \mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to c_3 (tfpt_4_frontier, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\text{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$, recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01 (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}}/r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$ is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\text{top}} = 48c_3^4$ that fixes the α -kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\text{rad}} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([C]); the amplitude $Q_e Q_m$ is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]:** the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi\nu\bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C]).** The closed TFPT CKM point ($s_{12}=\lambda_C$, $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$, $s_{13}=\lambda_C^3/3$, $\delta=\pi/3+3\lambda_C^2=1.19823$ rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_e, \kappa_\pm$) it gives $\text{BR}(K^+ \rightarrow \pi^+ \nu\bar{\nu}) = 9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu\bar{\nu}) = 3.33 \times 10^{-11}$ — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\text{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test [X]:** a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O]).** The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4 -character-block-diagonal iff f is $\mathbb{Z}_4$ -invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4 -mark sum $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \pmod{4}]$ for any profile g , hence is automatically $\mathbb{Z}_4$ -invariant ([E]). So the μ_4 -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conormal deck, v177)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered

in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver** (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]). The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170^\circ$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6\text{--}2.3$ (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120\text{--}130^\circ$ or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is $O(1)$ -uncertain, so a full finite- T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.
- **Full finite- T solve confirms it** (`axion_relic/full_finiteT_solve.py`). A converged nonlinear misalignment solve $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170^\circ$ — $\sim 5.5\times$ above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106^\circ$. The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199_seam_state_invariance.py** [E]/[O] (claim QGEO.STATE.01, work package A). Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4 -character classes $\{n=r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **v200_seam_variational_scan.py** [C]/[O] (claim QGEO.VARI.02, work package B). The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4 ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the F_{transfer} four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb_tfpt_v1.yaml) and its status semantics already exist.* Suite now 199 modules, highest ID v200; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198_modular_commutator_reduction.py** [O] (claim QGEO.MODULAR.01). The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$ and the clock $\rho: z \mapsto iz = \text{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω) , so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log \Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4 -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is μ_4 -invariant” ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in **tftpt_research_contracts**; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **v195_marks_lefschetz_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited): $\rho: z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component $\Rightarrow$ no puncture at a fixed point, and $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ D_4 marks” to the milder premise “ H^1 carries the A_3 -exponent rep” — less circular, still [O]. Cited in **tftpt_research_contracts**.
- **v196_seam_energy_functional.py** [E]/[O] (claim QGEO.VARI.01). The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the QGEO.SYM.01 conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4 -breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in **tftpt_research_contracts**.
- **v197_rr_carrier_clifford_d5.py** [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$. [E] arithmetic, [C] identification (as v189). Cited in **origin_theory**.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194_raw_seam_dtn_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \setminus \mu_4$. (3) [O] the residual relocates: the full- L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow

of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193_qgeo_energy_commutator.py [O] (claim QGEO.ENERGY.02).** The non-circular sharpening of QGEO.ENERGY.01: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, v117), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1, 2, 3)$ grading on H^1 (QGEO.COHOH.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by $5/3$) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **v189_riemann_roch_carrier.py [E]/[C] (claim ARCH.RRCAR.01).** The pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already QGEO.COHOH.01) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by rank(D_5) = 5. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **v190_nariai_entropy_bound.py [E] (claim HOR.NARIAI.02).** A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3}S_{\text{dS}}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191_universal_branch_line.py [C] (claim HOR.BRANCHLINE.01).** The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair $\{3, 4\}$ admits an equally exact map; the shared content is the known $2/3$ ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192_qgeo_energy_clock.py [O] (claim QGEO.ENERGY.01).** An operator-checkable restatement of the bedrock: $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). run_all.py green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **v184_etaB_anchored_boltzmann.py [C] (claim FR.ETAB.03).** An honest test of the external-review η_B proposal ($M_1 = M_R \varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{P1}$ both miss it (c_3/φ_0 -powers off $\sim 75\%/ \sim 40\%$), so $M_1 = M_R \varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in tfpt_4_frontier.
- **v185_axion_relic_solver.py [C] (claim FR.DM.03).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$ GeV ($m_a \approx 23.8 \mu\text{eV}$) CAN be all dark matter, but the harmonic misalignment under-produces ($\sim 21\%$ at $\theta \sim O(1)$), so the $\theta_i \approx 170^\circ$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in tfpt_4_frontier.
- **v187_ftransfer_laws.py [E] (claim FR.TRANSFER.01).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B , axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in tfpt_4_frontier.
- **v188_frontier_wording_guard.py [E] (claim AUDIT.WORDING.01).** A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording $\rightarrow$ v184 [C]; the old “open relic scale” dark-matter line $\rightarrow$ v185; the old “near, not exact” Koide line $\rightarrow$ v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the tfpt_4_frontier Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, QCD.LAMBDA.01).
- *Review framing points (P1/P2 as projections of the seam-deck postulate, v_{geo} as unit gauge, G_{net} a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183_koide_f_corner_transfer.py** [E]/[C] (**claim FR.KOIDE.07**). An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6 \times A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$) and $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$, where $F = R+Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M \mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source→pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [C], an F_{transfer} special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present**: the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, exponent ratio $|\mathbb{Z}_2|$, “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182_reviewer_residual_map.py** [E]/[C] (**claim REVIEW.MAP.01**). A careful external review raised four concerns (A mapping $2D \rightarrow 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ($\equiv \text{QGEO.SYM.01}$), D (Möbius forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck; the missing generator = F_{transfer}). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer} , v_{geo} , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ($\sin^2 \theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \leq 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate**. QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded**. Three rows (AX.P1.01, AX.P2.01, QGEO.IS0.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single

source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.

- **G_{net} typed in three explicit levels** (website “What is open” card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\text{fam}} \neq 3$); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox.** “no free fundamental number” softened to “no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~ 181 scripts (foundations $\rightarrow$ SM readouts $\rightarrow$ seam = horizon $\rightarrow$ horizon/flux geometry $\rightarrow$ R1–R5/premise-(A) $\rightarrow$ PyR@TE cross-checks $\rightarrow$ AQFT bridges $\rightarrow$ AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\text{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows `AX.P1.01/AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\text{car}} = e_2(a)$, bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181_clock_is_conformal_symmetry.py [E]/[O] (claim QGEO.SYM.01).** The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the μ_4 -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain **REALIZE(v175)** → $\dots$ → **SYM.01(v181)** ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the introduction “latest reductions” prose, **tfpt_5_redteam** Target A, the **tfpt_research_contracts** central-theorem keybox, **origin_theory**; and on the website **OpenGates**, **VerificationDag**, **papers.ts** (Target A $\times 2$, G_{net}), the **glossary** G_{net} entry and the **faq** live-residual answer — all now end at QGEO.SYM.01.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **v180_clock_is_mobius.py [E]/[O] (claim QGEO.ISO.01).** Cracks QGEO.CONF.01 one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, *non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z \mapsto iz$; (3) *isometry* $\Rightarrow$ *conformal* $\Rightarrow$ *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i , free orbit μ_4). Hence QGEO.CONF.01 is *implied* by the milder premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in **tfpt_research_contracts**; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **v179_conformal_realisation.py [E]/[O] (claim QGEO.CONF.01).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so “genus-0” is the same $\chi = 2$ that gives the “8” in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). KERNEL then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So

the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism”. The two doors of `v177/v178` are one geometric realisation, left honestly `[O]`. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations `v178` — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` `[E]/[O]` (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite `[E]` core of each and reduces each to *one sharper* premise. *Marks core* `[E]`: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ , the marks are forced to be a generic ρ -orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving a faithful D_4 of order 8 — so `MARKS` reduces to the single topological/clock premise. *Kernel core* `[E]`: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4 -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (`v162`) and the unique residue-normalised basis (`v177`) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So `KERNEL` reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/`v156`; no drift `v158`). Net: neither obligation closed; both sharpened to milder premises with their finite cores `[E]`. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree `v177` — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` `[E]/[O]` (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`). A second external residual-class audit pointed out that the `v176` statement takes the marks/ D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` `[E]` — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4 , and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful- D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening `v168`’s cross-ratio to the full uniformisation); `COHOM` `[E]` — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; `MODULE` `[E]` — a *unique* μ_4 -equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); `NET` `[E]` (`v154/v175`). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4 -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via `v168/v137/v162/v154/v175`). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem `v176` — the one central proof obligation)

- `v176_seam_collar_realisation.py` `[E]/[O]` (claim `QGEO.REALIZE.02`). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented

one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible $c=8$ data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1, 2, 3) = A_3$ exponents (v137/v168), μ_4 -equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$ (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \rightarrow \frac{53}{54}u$ source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's DEFAULT_RECORD_ID was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175_net_existence_full_cone.py [E]/[O] (claim QFT.NETRP.01).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m :* the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of $\text{spec}(t)$, so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \leq 6$, v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2) *A_2 assembled + verified:* the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 v156 check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174_seam_fock_readings.py** [E] (**claim QFT.FOCK.01**). Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four μ_4 -corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL ($\deg \leq 2$) keeps $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in **tfpt_2** (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170_diamond_slice_compression.py** [E] (**claim E8.SLICE.01**). E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3, 4, 6, 10)$ ($p_n = 2 + 2^n$) and Sheet-Diamond determinants $(3, 4, 8, 14, 20, 32)$ summing to $81 = N_{\text{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ follow; $78 = \dim E_6 = p_2(p_0 + p_3)$. Cited in **tfpt_3** (slice atlas). Exact; reading is audit (anti-numerology).
- **v171_os_moment_cluster.py** [E]/[C] (**claim QFT.OSMOMENT.01**). Atomic OS Moment Lemma: $\text{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster $729/665$, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$. Cited in **tfpt_research_contracts** (G5).
- **v172_trace_anomaly_seed.py** [E] (**claim SEED.ANOMALY.01**). The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 = \text{scalaron exponent } 48 - 41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ factorisation. Cited in **tfpt_2** (seed).
- **v173_pfaffian_cp_car.py** [E] (**claim FLAV.STRONGCP.02**). Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5 -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in **tfpt_2** (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo**. Version 5.1 (rev 97) of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release. Also fixed **build.sh zenodo** on macOS bash 3.2 (empty-array expansion under **set -u**).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **v168_qgeo_rigidity.py** [E]/[C] (claims QGEO.RIGID.01, QGEO.REALIZE.01). Hardens the *finite* half of the one remaining Tier-1 hinge GATE.QGEO (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1} dz / (z^4 - 1)$ ($k = 1, 2, 3$) are μ_4 -eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4 , four parabolic marks and this μ_4 -decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- **v169_etaB_boltzmann_interface.py** [C] (claim FR.ETAB.02). Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240^\circ$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10} , untuned) — the route is *viable*. But M_1 /washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **v166_higgs_free_seam.py** [E]/[C] (claim HIGGS.FREESEAM.01). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral $c=8$ fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from ~ 0.022 at one loop to ~ 0.002). The double condition predicts $m_H \sim 129\text{--}134$ GeV; the same condition at the scalaron scale gives ~ 107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ , reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- **v163_rg_stability_flavor.py** [E]/[C] (claim FLAV.RGSTAB.01). Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2 -suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2} Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$ and $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2 -driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **v164_qcd_scale.py** [E]/[O] (claim QCD.LAMBDA.01). The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop,

n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/ Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.

- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β ’s, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module v159_pyrate_gauge_crosscheck.py [E]/[C] (claim EM.BUDGET.02).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$, $\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}+1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.
- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_{transfer} + irreducible core, v167)

- **New module v167_inventory_post_closure.py [E]/[C].** The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (GATE.METRIC.16-19) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology $b_1=N_{\text{fam}}=3$); the EH step (R3) folds into v_{geo} (v152) and the ambient measure into G_{net} (v76). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer} , one functor with four typed conditional interfaces (Koide, η_B , axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger SYNTH.INVENTORY.03). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A_2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module v165_premise_a_closure.py [C]/[O].** Pins the two residual gates of the v160–v162 (A)-chain to their floor. A_2 (net existence) closes by assembly of estab-

lished mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)=16/16=1$, $248=120+128$, character $E_4/\eta^8=j^{1/3}$ (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via $DtN = |k|$ (v156/v157). R_1 (**seam clock**) **reduces to GATE.QGEO** [C]: a μ_4 -equivariant flat gapped generator is *unique* ($= A_0^*$, v115/v116), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification “seam boundary $= \mathbb{P} \setminus \mu_4$ ” = **GATE.QGEO** (cohomology established, v137, $b_1=3=N_{\text{fam}}$). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay $\{\pi, v_{\text{geo}}\}$; premise (A) = (A_2 assembly) + ($R_1=\text{GATE.QGEO}$) + the $\{\pi, v_{\text{geo}}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger **GATE.METRIC.19**). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module v162_seam_transport_identification.py** [E]/[O]. Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4 -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$ for $U = \text{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (v115) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2=6=|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of $\omega_k=|k|$ (v113/v156). **Closure:** with v160+v161, (A) closes given only μ_4 -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, **HOR.CLOCK**). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger **GATE.METRIC.18**). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, v161)

- **New module v161_one_particle_reduction.py** [E]/[O]. *Discharges* the scope premise of v160 (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree- m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\text{spec}(\Gamma(t)|_{\text{deg } m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all $m=0..6$); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation $\Leftrightarrow$ one vacuum); (4) quasi-freeness is *forced* — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ , gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo

class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger GATE.METRIC.17). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}=0$ — verified by the *signed* set-partition/Pfaffian moment $\leftrightarrow$ cumulant inversion on a concrete pure Majorana covariance (the v113 “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray $\Rightarrow$ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap}>0$, the value $(2/3)^6$ only sets the rate. **The honest residual:** v56’s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}=1820$ and $\binom{16}{6}=8008$ higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.
- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection \times \{YYYY-MM-DD...\}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral $c=8$ fixed point is provably stable (v158, exact dimension count).** The finite core of (A)’s “reaches-the-fixed-point” residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current $h=1$, the quartic $h=2$. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the $h=1$ currents are chiral (v157, not $(1,1)$ -marginal), and the lowest interaction (quartic, $h=2$) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the $b_1=41/10$ cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; `contracts / origin_theory / next.txt / website` moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*

$|k|$ (Lee–Uhlmann — order-1 Ψ DO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic $c=8$ theory is *rigid* — every field has $\bar{h}=0$, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving $\mu=1$ ($248=120_{\text{NS}}+128_{\text{R}}$). With $(E_8)_1$ unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ($\Delta=6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).

- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5+3$; (c) the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q -series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual*: exact given (i) the free-bulk premise (DtN = $|k|$, v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger GATE.METRIC.13).
- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN = $|k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the $248=120+128$ split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- G_{net} ’s last premise reduces to a single shared physical premise (v155, Python-only). The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression PTP of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by G_{net} , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger GATE.METRIC.12).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- v_{geo} **re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. v_{geo} is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals

collapse onto it ($U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay {one unit, π } (ledger `ANCHOR.VGEO.02`).

- **G_{net} stated as the central closing theorem (v154).** The Simple-Current Extension Theorem: $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has $[B:A]=|L|=4$, $c=5+3=8$, $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed** (review): the canonical status card now reads v_{geo} as a metrology *primitive*, G_{net} as the simple-current extension $\cong (E_8)_1$ ($[\mathbf{E}]/[\mathbf{C}]$), and F_{transfer} as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\text{geo}}\}$; one seam-net theorem; frontier = F_{transfer}) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/origin_theory/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts $\leftrightarrow$ papers $\leftrightarrow$ ledger $\leftrightarrow$ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d=2$ that ratio *is* the induced Newton constant $1/G$ (v68). So v_{geo} (v78), $1/G$ (v68) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π }, and `SEAM.THEOREM.01`’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin_theory/next.txt/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO_TOKEN secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to `main`, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ~270 files / ~3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references; stale literal [A]/[I]/[L] markers in running text replaced by the 4-class [O]/[E].

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references in all three documents; the horizon “Scope/typing” marker line and a stale literal [A] in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` on *failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón

jump determinant *vanishes identically* ($C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by v150 needs no separate derivation. SEAM.THEOREM.01 narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays [O] (ledger SEAM.EHMODEL.02).

- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin_theory/next.txt moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4': both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2 \cdot 1$ the dual normal pair is anchor data through and through. Equivalences to the $(2, 8, 121)$ frame data and the w_0 -lift exact. R4' residue = *one* discrete assignment (ledger GATE.UWALL.14).
- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and t -independent (Cheeger image sum $(N^2 - 1)/(12N)$); for a *gapped* operator the ζ -regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma) \ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m / 12\pi) \int \sqrt{g} R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, origin_theory, contracts, next.txt, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma) + |\mu_4|e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 (4+5=9, \sigma \perp w_0(a))$ and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{PI}(K)\|_1$. R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z \mapsto iz, z \mapsto 1/z \rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity +1 on the self-conjugate line (= T_A in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is $\det'$ -clean (v147).** Variance ratio $(1 - \alpha)$ (forced by norm = area) gives the Born-squared ratio $(1 - \alpha)^{p_2}$, the $\ln$ series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\text{NS}} + 128_{\text{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum $q_D + q_A$ (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.

- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{P1}$ vs M_{P1} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies” $\rightarrow$ “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).
- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, $2/3$ motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + $c=8 \Leftrightarrow$ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net} : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A \Sigma$ (v97/v98) pairs the $Q_+=1$ cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2, 3, 1) = \text{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \rightarrow -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading = $Q_+ \circ \rho$. GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings $\rightarrow$ two (v142, frame integrality).** Integer covectors in the $(1, a, \sigma)$ dual frame form an index-11 sublattice ($x - 3y + z \equiv 0 \pmod{11}$; the index is $\|\text{Pl}(K)\|_1$);

with $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$ the σ -pairing is forced $\equiv 0 \pmod{11}$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger `GATE.UWALL.12`).

- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$ -graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger `GATE.METRIC.09`).
- **R1’s det-ratio step derived in-family (v144).** e_2 -rigidity of the traceless SdS cubic gives $r_{br_c} = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays `[C]` with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply (ledger `HOR.CLOCK.12`).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant $\rightarrow$ EH step (`SEAM.THEOREM.01`); the gate keeps its `[O]` typing untouched.
- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (`GATE.WOLFRAM.02` and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script $\rightarrow$ check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite $\leftrightarrow$ `run_all.py` $\leftrightarrow$ registry $\leftrightarrow$ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.

- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal [P] in `tfpt_4` (Koide flow-time) replaced by the 4-class [C]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [E]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ($N_\star=51.4 \Rightarrow n_s=0.9611$, $r=0.0045$, A_s -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [C]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [E] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [C] conditional (physical / bridge / readout under named hypotheses), [O] open / axiom, and [X] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations

(e.g. $[I]/[L] \rightarrow [E]$, $[N]/[P] \rightarrow [E]/[C]$). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.

- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$ rows are made consistent with `v86/COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star=51.4$ ($n_s=0.9611$, $r=0.0045$), replacing the stale point values $N_\star=55$ (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed $[E] \rightarrow [C]$ (conditional on $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$), matching the other two tables and the ledger $[P]$ typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry author, date and `v5.2 (rev N)` automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. `v108–v113`, `v49/v71`) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (`v1`, `v6/v14/v23`, `v18/v20/v46`).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`).

`changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.

- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long `vN`→check table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven **keyboxes** are reduced to the single headline “In one sentence” card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a **winbox** (a result, not a claim), and the explanatory *plausible/follows next* **winboxes** become prose. The loud boxes now carry only genuinely load-bearing callouts (one **keybox**, five **winboxes**, one **gapbox**) plus the two shared orientation cards.
- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only **keybox** (claim), **winbox** (result) and **gapbox** (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex` (`\input` by `tfpt_style`).** reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the $2/3$ *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), **[C]** (readout / standard physics in TFPT units) to the existing **[C]** (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are **[E]** closed on the derived stratum, only the absolute scale stays **[O]**, with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (leading c_3 , fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result (0.3092 ± 0.0087 , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence

hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.

- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ (historical $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$); F_{transfer} is named as one missing functor with Koide/ η_B /axion/ m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv \text{tfpt_N}$). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \rightarrow 12 \rightarrow 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ($\text{spec } A_0^*$); $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law $\rightarrow$ zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).

- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1, 1, -2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\text{rowsum}(V) = \text{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the A_0^* zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the A_3 exponents and cusp weights $= \text{spec } A_0^*$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the $R4'$ residue is three pairings; the $H^1 \rightarrow$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the GATE.QGEO residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ~ 110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tftpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s, t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, $\text{rank} = c$) — the QBL input merges with the $R2$ /holomorphy premise; communicated across all status surfaces.
- **U_{wall} made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue A_0^* ($\text{diag} = a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff$

$a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.

- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.